

EXPERIENCE

- **TP-Link** Irvine, CA
Embedded Software Engineer *July 2025 - present*
 - **OpenWrt**: embedded system development in C using buildroot and OpenWrt for TP-Link SMB gateway products
 - **ML & Arm TrustZone**: Assigned lead role in exploring usage of the machine learning (on/off device) and Arm TrustZone in relation to that
- **Freelance Software Engineer** San Diego, CA
Software Engineer *Sept 2024 - July 2025*
 - **NVIDIA Jetson AGX Orin Dev Kit**: Implemented secure boot for AI startup and used LXC containers for on device sandboxing
 - **AI Web form Automator**: Created a chrome extension for a client to upload a PDF file and automatically fill in a web form
- **Qualcomm** San Diego, CA
Software Engineer - Secure Systems Group *May 2022 - Dec 2023*
 - **C Library**: Built a library in C for use on Embedded Linux and Android. Involved concepts used for IPC.
 - **Trusted Virtual Machine Development**: Integrated the C Library I built and updated the unit testing suite used to test the Trusted Virtual Machine
 - **Test Driven Development**: Designed C++ code with a Test Driven Development approach using the Google Test framework
 - **Static Analysis**: Fixed code security vulnerabilities for multiple repositories using an internal static analysis tool to audit code
- **Intel** Chandler, AZ
Software Engineer - Graduate Student Intern *Nov 2019 - Dec 2021*
 - **12x Optimization**: Designed and implemented a 12x data processing optimization (using multithreaded code and a lighter library) to reduce the run time from 1 hour to under 5 minutes. (C#)
 - **Group Presentation**: Gave a presentation to a group of software engineers on development tools and potential uses for the group: Docker, GitLab CI/CD
 - **Docker**: Designed and implemented a replica database using Docker to use for development (previously the group has used the production database during testing phases)
- **Arizona State University** Tempe, AZ
Graduate Student Assistant - CSE 539 Applied Cryptography *Aug 2021 - Dec 2021*
 - **Course Design**: Proposed, designed, and implemented extra credit projects for the students: Time-Based One-Time Passwords (TOTP) in Python to match the Google Authenticator app on their phones.
 - **Projects**: Developed correct solutions for the five major projects in the course: Steganography, cryptanalysis, MD5 hash collisions, Diffie-Hellman key exchange, and RSA key exchange (C#)

EDUCATION

- **Arizona State University** Tempe, AZ
Master of Science in Computer Science *Aug. 2019 - Dec. 2021*
- **Brigham Young University** Provo, UT
Bachelor of Science in Physics *2017*

PROJECTS

- **Virtual Computer**: I built a virtual computer following the Nand2Tetris course offered by the Hebrew University of Jerusalem

HONORS/AWARDS

- **Hackathon 2nd Place Team**: I was the lead software engineer on my team. I built a basic LeNet convolutional neural network using Keras in Python. Hosted by FOX at ASU.

PROGRAMMING SKILLS

- **Languages**: C++, C, Python, Rust **Technologies**: Linux, Nix, NixOS, GDB, Gerrit, GitHub